Draft the SQL migration for the canonical partner mapping

Show how to separate billing accounts from marketplace partners
Design the transaction classification table for GMV tracking
Propose the isolated staging environment setup for testing
Review the next reviewable artifact for deployment approval

Drafted two reviewable artifacts: the marketplace canonical-mapping SQL migration and an isolated-staging/deployment review checklist.

The design keeps **marketplace identity separate from billing identity**, adds explicit transaction classification for GMV, and fails closed for browser access. Nothing was applied to Supabase, deployed, or scheduled. The SQL is an untested review draft—not an approved production migration.

1. Separate billing accounts from marketplace partners

A billing account answers:

> Who is responsible for charges, invoices, and billing administration?

A marketplace partner answers:

> Who supplies the eligible goods or services represented in marketplace transactions?

These are related but not interchangeable.

Proposed relationship

```
Existing canonical organization/person
                 |
       Marketplace-role identity
                 |
          Marketplace partner
                 |
       Approved billing association
                 |
       Existing billing account
                 |
        Invoices / subscriptions
```

The canonical organization link remains a review blocker: the inspected schema did not establish which record represents commercial partner organizations. The draft therefore does not reuse student organizations or integration providers as sellers, nor create a second legal-entity master.

Why an association table is needed

Examples:

- One partner can use different billing accounts for distinct approved purposes.
- A billing account can serve multiple partner relationships.
- A partner may transact before it has a billing arrangement.
- An invoice may be for platform subscriptions rather than marketplace activity.
- A billing-account link may expire without the partner ceasing to exist.

The mapping must record purpose, effective dates, source identity, and authorization evidence. It must not itself grant access.

Your existing architecture explicitly warns against introducing parallel canonical models or permission predicates. This migration should extend that foundation after reconciliation. [1]

2. What the SQL draft creates

All five names below are proposed additions, not assertions that these tables already exist.

Proposed table	Grain	Purpose
`marketplace_partners`	One source-attributed partner identity within a school scope	Establish marketplace role identity
`marketplace_partner_billing_links`	One approved effective-dated association	Connect a partner to an existing billing account
`marketplace_transactions`	One source-attributed transaction	Preserve eligible financial inputs and provenance
`marketplace_transaction_classifications`	One classification revision per transaction	Determine GMV eligibility under a versioned policy
`marketplace_transaction_adjustments`	One source financial event	Record refunds, commissions, reversals, and variable costs

The draft uses only the two previously verified foreign-key column types:

- `public.schools.id`: text.
- `public.billing_accounts.id`: UUID.

Other relationships require inspection before executable deployment approval.

Mapping excerpt

```
CREATE TABLE public.marketplace_partner_billing_links (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  school_id text NOT NULL,
  partner_id uuid NOT NULL,
  billing_account_id uuid NOT NULL
    REFERENCES public.billing_accounts(id),

  purpose text NOT NULL CHECK (
    purpose IN (
      'fee_billing',
      'payout_reference',
      'program_sponsorship'
    )
  ),

  valid_from timestamptz NOT NULL,
  valid_until timestamptz,

  authorization_reference text NOT NULL,
  source_namespace text NOT NULL,
  source_link_key text NOT NULL,
  recorded_at timestamptz NOT NULL DEFAULT now(),

  FOREIGN KEY (school_id, partner_id)
    REFERENCES public.marketplace_partners(school_id, id),

  CHECK (
    valid_until IS NULL OR valid_until > valid_from
  ),

  UNIQUE (
    school_id,
    source_namespace,
    source_link_key
  )
);
```

A `payout_reference` link is only a reference. It does not demonstrate payment capability, establish custody, or authorize money movement.

Important unresolved mapping controls

Before writes are enabled:

- Verify that the billing account is authorized for the intended scope.
- Define whether associations may overlap.
- Define revocation and amendment behavior.
- Bind approval evidence to the exact association.
- Integrate existing capability checks and audit.

- Verify the canonical business-organization relationship.

The draft intentionally does not invent these controls.

3. Transaction classification for GMV

An amount becomes GMV only after its commercial meaning and eligibility are established.

Classification dimensions

Field	Examples	Why it matters
Commercial classification	Marketplace merchandise, platform subscription, donation, deposit, tax-only, unclassified	Prevents billing activity from inflating GMV
Fulfillment state	Pending, fulfilled, cancelled, failed, not applicable	Establishes eligibility timing
Inclusion decision	Included, excluded, review	Keeps uncertain records out of confirmed reporting
Policy version	Approved metric-policy identifier	Makes results reproducible
Evidence reference	Source transaction or reviewed document	Supports reconciliation
Decision authority reference	Authorized classification decision	Preserves accountability
Revision	Increasing per-transaction version	Supports corrections without deleting history

Inclusion rule

The draft permits `included` only when:

```
classification = marketplace_merchandise
AND fulfillment_state = fulfilled
AND fulfilled_at is present
```

Examples:

Transaction	GMV treatment
Fulfilled eligible marketplace service	Include under the approved policy
Semester platform subscription invoice	Exclude
Admission deposit	Exclude from this marketplace metric
Donation	Exclude
Cancelled marketplace order	Exclude
Ambiguous invoice line	Review; do not silently include

These are proposed metric-policy choices, not approved accounting policy.

Amounts and currencies

The draft stores:

```
merchandise_minor
discount_minor
tax_minor
shipping_minor
currency
```

Proposed gross GMV:

[
Gross GMV

Merchandise amount

Seller discount
]

Tax and shipping remain separately recorded but excluded from that definition.

The SQL uses currency minor units—not universally “cents.” The ingestion adapter must validate the currency’s exponent before formatting amounts.

Adjustments and safe aggregation

Refunds, commissions, commission reversals, and variable costs are separate events.

```
Transaction
 ├─ Merchandise refund events
 ├─ Commission events
 ├─ Commission reversal events
 └─ Variable-cost events
```

The companion query aggregates adjustments per transaction before joining them. This avoids multiplying fees and refunds through a many-to-many join.

It also selects the latest classification recorded by a specified cutoff, making a historical report reproducible.

4. Security posture of the draft

The migration:

- Enables RLS on all five proposed tables.
- Grants no browser access.
- Creates no browser policies.

- Adds no new authorization predicate.
- Gives the trusted service role only `SELECT` and `INSERT` on these tables.
- Uses composite foreign keys to prevent cross-school partner/transaction references.
- Adds unique source keys for duplicate protection.
- Creates no cron job or Edge Function.

However, service-role operations bypass RLS. Trusted ingestion must therefore validate scope, permissions, classification authority, source currency, and financial limits before writing.

Operational append-only grants also do not establish absolute immutability: database owners can still change data. Do not claim immutable audit protection until ownership, audit integration, and reviewed command paths are tested.

The reference migration is deliberately fail-on-collision. Adding `IF NOT EXISTS` could hide a divergent existing schema.

5. Isolated staging setup

The recommended test environment is a dedicated development branch or separately approved staging project—not the production database and not the existing restore-drill project.

Environment plan

Component	Staging requirement
Database	Reviewed baseline migrations plus this draft
Fixtures	Synthetic schools, billing accounts, partners, transactions
Application	Dedicated preview configuration pointing only to staging
Credentials	Separate test credentials; no production service keys
Payments	Sandbox adapters or fixtures; no real settlement
Notifications	Test outbox; no real email/SMS delivery
AI	Disabled unless specifically required for a reviewed test
Cron	Disabled until worker and alert tests pass
OAuth/webhooks	Staging callbacks and endpoints only
Reporting	Clearly labeled synthetic data and unapproved metrics

Provisioning may incur cost. Any branch/project creation requires a cost check and separate confirmation before execution.

This follows the project's migration pattern: discovery and mapping precede staging, validation, reconciliation, and cutover. [2]

Required test cases

- Migration applies against the reviewed baseline.
- Unexpected table collisions fail visibly.
- Browser roles cannot read or write the new tables.
- Cross-school references are rejected.
- A billing link cannot bypass account-scope authorization.
- Duplicate source events are rejected.
- Conflicting replay payloads are quarantined by ingestion.
- Subscription, deposit, tax-only, cancelled, and ambiguous records do not enter confirmed GMV.
- Multiple adjustment events do not produce join fanout.
- Historical cutoff reports reproduce the expected classification.
- Currency mismatches and excessive refunds/reversals are rejected.
- Missing lifecycle or ROI inputs remain unavailable.
- Incomplete reporting periods block business alerts.
- Revocation, amendments, and concurrent ingestion behave safely.

6. Deployment review: current verdict

Review area	Status
Billing versus partner separation	Drafted
Transaction grain and provenance	Drafted
Classification and revision model	Drafted
Adjustment aggregation	Drafted
Existing organization mapping	Blocked pending inspection
Billing-account scope authorization	Blocked pending inspection
Capability and audit integration	Blocked pending implementation review
Partner lifecycle, tiers, ROI evidence	Separate follow-on design required
Staging execution	Not performed
SQL/database tests	Not performed
Worker scheduling and notification	Not created
Production approval	Not ready

The next approval artifact should be a staging-only packet containing:

1. Exact migration diff and commit.
2. Confirmed target environment.

3. Canonical organization and scope mapping.
4. Permission, audit, revocation, and ingestion design.
5. Synthetic fixtures and expected calculations.
6. Test results.
7. Finance-approved metric contract.
8. Rollback/recovery procedure.
9. Named engineering, finance, security, and operating owners.

For rollback, first disable consumers and ingestion, preserve evidence, and use a reviewed corrective migration. Do not automatically drop populated financial-attribution tables.

The attached SQL and review checklist are ready for technical review. They are not yet ready for deployment approval because the canonical organization relationship and scoped billing authorization remain unresolved.

⁂

1. The Main Semester design system (2) copy/docs/system/ROLE_SYSTEM_RECONCILIATION.md
2. The Main Semester design system (2) copy/docs/platform/INTEGRATION_AND_MIGRATION_FACTORY.md
3. The Main Semester design system (2) copy/docs/strategy/SEMESTER_MIGRATION_FACTORY.md
4. The Main Semester design system (2) copy/handoff/control-plane/baseline/db/001_access_jobs.sql